

CURBCOUNT · REQUIREMENTS

CurbCount requirements

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DOCUMENT CONTROL

Document ID	CBC-REQ-001	Version	0.3
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/curbcount		
Source	docs/03-requirements.md @ 493c9b3		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from CBC-CAL-001; design case bike lane 2.0 m to match R6; decisions from CBC-DDR-001 reflected; proposed restatements of R4 and R11 noted
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (numbered, measurable requirements with targets and concept status)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CurbCount requirements

These requirements are targets for concept review and must be revised with a co-design partner before the design is frozen. The design choices behind them (thermal array, high-load FieldNode power, ESP32-S3 processor, 15-minute bins and three classes, jumper-only calibration, public notice) are adopted for TRL 3 under Amish's 2026-09-25 instruction and open for his review (CBC-DDR-001). Status is taken from the calculation note CBC-CAL-001; "met on paper" means met by calculation only. Three requirements are **not met**: R4 as written (the privacy aim is met), R7 in hot weather and R13 cost. Four are at risk (R1, R2, R11, R15) and four cannot be shown at TRL 3 (R3, R10, R12, R14).

Design case: one counter on a 114 mm pole 0.45 m behind the curb, sensor window 4.3 m above the road, a 3 m sidewalk, a 2.0 m bike lane and one 3.5 m traffic lane. (Version 0.2 used a 1.8 m bike lane; 2.0 m matches the R6 target.)

Table 1. Requirements.

ID	Requirement	Target	Verification	Status (CBC-CAL-001)
R1	Count pedestrians, cyclists (including micromobility) and motor vehicles in the nearest lane, by direction along the street	3 classes x 2 directions	Test against manual counts	At risk: 4.2 frames for a car at 50 km/h, 4 assumed needed
R2	Count accuracy for pedestrians and cyclists	Within ± 10 % of manual counts per hour at flows up to 600 per hour per direction	Field comparison with manual counts	At risk: random merging alone loses 6.5 %; groups add more
R3	Count accuracy for motor vehicles in the nearest lane	Within ± 15 % of manual counts per hour	Field comparison	Not verifiable at TRL 3 (4.2 frames, 3.4 px smear)
R4	Privacy: no image, audio or identifier leaves the device; frames held only in RAM; sensor unable to resolve faces or plates	Ground pixel size 0.25 m or larger at the design height	Design review, firmware audit, geometry	Not met as written: smallest ground pixel 0.22 m. Privacy aim met: 8.5 px/m at head height against 250 px/m to identify (restatement proposed, awaiting Amish)
R5	Output format	15-minute counts per class and direction in an open, documented format over LoRaWAN	Design review	Met by design (14 bytes per bin)
R6	Coverage from one pole	Full width of a 3 m sidewalk, a 2 m bike lane and the nearest lane	Geometry, field check	Met on paper with the array turned 90 degrees: -4.45 to 8.34 m across
R7	Operating range	-20 °C to +50 °C; counting works with air up to 35 °C	Thermal chamber, hot-season field test	Not met on paper: on a hot design day pedestrians fall below detection contrast for 14.2 h (sunlit) to 24 h (shaded)
R8	Autonomy without sun	3 days of continuous counting	Energy budget, bench test	Met on paper: 4.78 days with two cells (3.35 at -20 °C); 2.39 with one
R9	Energy neutral in winter	At 1.5 peak sun hours with street shading derating	Energy budget, field log	Met on paper: 20 W stores 14.5 Wh/day against 6.43 Wh/day; 6 W does not
R10	Ingress protection	IP65 for the enclosure and the sensor head	Spray test	Not verifiable at TRL 3
R11	Mass on the pole	6 kg or less	Weighing	At risk: 6.02 kg (TRL 2 estimate was 4.2 kg)
R12	Installation	Two trained people, 60 min, no drilling, welding or pole wiring; withstands 35 m/s gusts	Timed trial, bracket calculation	Not verifiable at TRL 3: 59 min estimate; wind and fit met on paper
R13	Parts cost	\$150 or less per counter (budget of \$275)	Priced BOM	Not met: \$290.00

ID	Requirement	Target	Verification	Status (CBC-CAL-001)
		recommended, awaiting Amish)		
R14	Service life	5 years outdoors with one battery change	Design review, UV-stable materials	Not verifiable at TRL 3
R15	Radio use	Within EU868 1 % duty cycle and the US915 dwell limits	Airtime calculation	At risk: 0.91 s/h at SF9; the 14-byte record exceeds the 11-byte US915 DR0 limit

Table 2. Assumptions behind the targets.

Assumption	Value	Source or basis
Walking speed	1.0 to 1.5 m/s	Typical design range (estimate)
Cyclist speed	up to 25 km/h	Typical urban maximum (estimate)
Vehicle speed in the nearest lane	up to 50 km/h	Common urban limit (estimate)
Peak sun hours, winter design case	1.5 h	Mid-latitude winter (estimate)
Street shading and dust derating	40 %	Estimate, to be checked on real poles

Proposed changes awaiting Amish (see `docs/REVIEW.md`): restate R4 as "8 px/m or less at head height, below the IEC 62676-4 detection level of 25 px/m" in place of the ground pixel size; keep R11 at 6 kg and trim mass, or relax it to 6.5 kg; set the payload for R15 at 10 bytes.